

Joint Identification of High-dimensional Interaction Networks Based on Longitudinal Data with Applications to Vaccine Immunization

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Abstract

Motivated by the vaccine immunization study, joint identification algorithms for interaction networks corresponding to the pre- and post- vaccination is proposed in this paper. The algorithms aim to improve the traditional methods through directly modeling the correlation structure between longitudinal high-dimensional observations which is common in vaccine study. The algorithms are grouped into two classes in terms of the homogeneity of the subjects. Simulation studies are carried out which shows that the proposed algorithms outperformed the existing ones when the correlation between different time points are reasonably large.

1. Introduction

Exploring the interactions between different species has become the central theme of systemic biology, such as interactions between genes, proteins, antibodies and microbial strains *et al.* Traditionally, the interactions are studied in lab using co-culture assays under controlled conditions, such as pH, oxygen levels, and nutrient availability (Gause (1934); Harcombe (2010)). With the advance of high-throughput technologies, omics data becomes more and more common in which hundreds of species are measured from the same specimen. It becomes realistic to build the interaction networks based on such omics data.

With omics data, many different statistical models have been proposed to build the interaction networks. The simplest yet the most common way is based on the Pearson correlation coefficient (Qin et al (2010); Barberan (2012)). Compared to the Pearson correlation, Gaussian graphical models (Friedman et al (2008b)) can eliminate the possible confounding and have become increasingly popular for inferring the interaction network (Kurtz et al (2015); Chen et al (2017); Tian et al (2023)).

In clinical research settings, many omics data are longitudinal. The correlation between omics data could have a big impact on the estimation of the networks. To handle the correlation, time series model such as vector autoregression have been used to identify the networks (Epskamp (2018); He et al (2022)). However, time series models are not able to handle the general longitudinal omics data where the data usually are spaced irregularly. In particular, when the underlying process of longitudinal data is stationary, Zhou et al (2024) proposed longitudinal graphical lasso model to estimate the microbial interaction networks based on 16S RNA sequences.

In this paper, we consider the networks estimation from the longitudinal data during which a treatment is administered to each subject. For example, in order to study the effectiveness of a new vaccine, the antibodies are measured multiple times before and after the vaccination for each subject in the cohort. Since it is believed that the vaccination changed the structure of interaction

network of antibodies, it is desirable to estimate both the pre- and post-vaccination networks. It should be pointed out that statistical models have been designed in literature to estimate multiple networks simultaneously (Danaher et al (2014); Gao et al (2016); Hao et al (2018); Ren et al (2021, 2022)). However, none of these models can handle the longitudinal data with treatment.

Based on the work in Zhou et al (2024), we proposed a two-stage longitudinal Gaussian graphical models which aim to estimate the pre- and post-treatment networks simultaneously from longitudinal omics data. We demonstrate the effectiveness of the model using simulated data. The models are then employed to estimate the antibody interaction networks before and after the vaccination.

2. A statistical model for longitudinal antibody titers

Let $y_{itk} = (y_{itk1}, \dots, y_{itkp})^T$ denote the titers of p antibody features for subject i at time point k ($1 \leq i \leq m$, $1 \leq k \leq n_i$). The vaccine is administered at some time point $n_{i(1)}$ so that before the vaccination, we have $y_{itk} \sim N_p(\mu_1, \Sigma_1)$, while after the vaccination, we have $y_{itk} \sim N_p(\mu_2, \Sigma_2)$, where $\mu_1 = E(y_{itk})$, $\Sigma_1 = \text{Var}(y_{itk})$ for $1 \leq k \leq n_{i(1)}$, and $\mu_2 = E(y_{itk})$ and $\Sigma_2 = \text{Var}(y_{itk})$ for $n_{i(1)} < k \leq n_{i(1)} + n_{i(2)} \triangleq n_i$. The precision matrices before and after the vaccination are defined as $\Omega_{(1)} = \Sigma_{(1)}^{-1}$, and $\Omega_2 = \Sigma_2^{-1}$ respectively. The data from subject i at all n_i time points are stored in the $n_i p$ vector $y_i = (y_{it_1}^T, \dots, y_{it_{n_i}}^T)^T \triangleq (y_{i(1)}^T, y_{i(2)}^T)$, where the $n_{i(1)} p$ vector $y_{i(1)}$ and $n_{i(2)} p$ vector $y_{i(2)}$ represents the data before and after the vaccination respectively. In order to facilitate the computation in the next section, we format the data in the following matrix form. Specifically, let $n_{i(1)} \times p$ matrix Y_{i1} and $n_{i(2)} \times p$ matrix Y_{i2} be the pre-vaccination and post-vaccination data matrices respectively, where each row of Y_{i1} and Y_{i2} is the measurement at a given time point for subject i . The full data matrix for subject i is thus a $n_i \times p$ matrix $Y_i = (Y_{i(1)}^T, Y_{i(2)}^T)^T$. This way, we have $y_i = (\text{vec}(Y_{i1}^T)^T, \text{vec}(Y_{i2}^T)^T)^T \triangleq \text{vec}(Y_i^T)$ where the symbol vec is the vectorization operator which stacks the columns of a matrix one after another. Then we pool all y_i 's together and have vector $y = (y_1^T, \dots, y_m^T)^T = (\text{vec}(Y_1^T)^T, \dots, \text{vec}(Y_m^T)^T)^T \triangleq \text{vec}(Y^T)$, where $n \times p$ matrix Y represents the full data set from all subjects at all time points in which $n = \sum_{i=1}^m n_i$. Without loss of generality, we always assume that $\mu_1 = \mu_2 = 0_p$. Our goal is to estimate the high-dimensional precision matrices, i.e., the networks Ω_1 and Ω_2 simultaneously based on the full data set y .

To this end, for $y_{i(1)}$, the data set of subject i before the treatment, we assume,

$$\begin{aligned} \text{Var}(y_{i(1)}) &= \begin{pmatrix} \Omega_{(1)}^{-1} & \rho_{i(1)12}\Omega_{(1)}^{-1} & \cdots & \rho_{i(1)1n_{i(1)}}\Omega_{(1)}^{-1} \\ \rho_{i(1)21}\Omega_{(1)}^{-1} & \Omega_{(1)}^{-1} & \cdots & \rho_{i(1)2n_{i(1)}}\Omega_{(1)}^{-1} \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{i(1)n_{i(1)}1}\Omega_{(1)}^{-1} & \rho_{i(1)n_{i(1)}2}\Omega_{(1)}^{-1} & \cdots & \Omega_{(1)}^{-1} \end{pmatrix} \\ &= \begin{pmatrix} 1 & \rho_{i(1)12} & \cdots & \rho_{i(1)1n_{i(1)}} \\ \rho_{i(1)21} & 1 & \cdots & \rho_{i(1)2n_{i(1)}} \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{i(1)n_{i(1)}1} & \rho_{i(1)n_{i(1)}2} & \cdots & 1 \end{pmatrix} \otimes \Omega_{(1)}^{-1} \triangleq \Phi_{i(1)} \otimes \Omega_{(1)}^{-1} \end{aligned} \quad (1)$$

where $\otimes$ represents Kronecker product, and matrix $\Phi_{i(1)}$ represent the correlations between different time points for subject i before the vaccination. Similar to Zhou et al (2024), among many other options, we adopt the following parameterization for $\Phi_{i(1)}$,

$$\rho_{i(1)j_1j_2} = \exp(-\tau_{i(1)}|t_{j_1} - t_{j_2}|^k) \quad (2)$$

which has been widely employed in the literature.

Similarly, for $y_{i(2)}$, the data set of subject i before the treatment, we assume,

$$\begin{aligned} Var(y_{i(2)}) &= \begin{pmatrix} \Omega_{(2)}^{-1} & \rho_{i(2)12}\Omega_{(2)}^{-1} & \cdots & \rho_{i(2)1n_{i(2)}}\Omega_{(2)}^{-1} \\ \rho_{i(2)21}\Omega_{(2)}^{-1} & \Omega_{(2)}^{-1} & \cdots & \rho_{i(2)2n_{i(2)}}\Omega_{(2)}^{-1} \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{i(2)n_{i(2)}1}\Omega_{(2)}^{-1} & \rho_{i(2)l_i2}\Omega_{(2)}^{-1} & \cdots & \Omega_{(2)}^{-1} \end{pmatrix} \\ &= \begin{pmatrix} 1 & \rho_{i(2)12} & \cdots & \rho_{i(2)1n_{i(2)}} \\ \rho_{i(2)21} & 1 & \cdots & \rho_{i(2)2n_{i(2)}} \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{i(2)n_{i(2)}1} & \rho_{i(2)n_{i(2)}2} & \cdots & 1 \end{pmatrix} \otimes \Omega_{(2)}^{-1} \triangleq \Phi_{i(2)} \otimes \Omega_{(2)}^{-1} \end{aligned} \quad (3)$$

where matrix $\Phi_{i(1)}$ represent the correlations between different time points for subject i after the vaccination. Similarly, for correlations between different time points, we assume

$$\rho_{i(2)j_1j_2} = \exp(-\tau_{i(2)}|t_{j_1} - t_{j_2}|^k) \quad (4)$$

As for the correlations between different subjects, we assume the independence, i.e.,

$$Cov(y_{i_1k_1}, y_{i_2k_2}) = 0_{p \times p} \quad (5)$$

for $1 \leq i_1 \neq i_2 \leq m, 1 \leq k_1 \leq n_{i_1}, 1 \leq k_2 \leq n_{i_2}$.

Based on whether or not subjects share the same correlation parameters, the models above can be divided into two categories. In first category, all subjects share the same correlation parameters and hence are called homogeneous model while in second category subjects have their own correlation parameter and hence are called heterogeneous model.

2.1. Homogeneous models

In homogeneous models, it is assumed that $\tau_{1(1)} = \cdots = \tau_{m(1)} = \tau_{(1)}$ and $\tau_{1(2)} = \cdots = \tau_{m(2)} = \tau_{(2)}$, which combined with the normal distribution assumption yields the following log-likelihood function of $y_{(1)}$,

$$l_n(\Sigma_{(1)}, \tau_{(1)}|y_{(1)}) = -\frac{1}{2} \sum_{i=1}^m \left(p \log(|\Phi_{i(1)}(\tau_{(1)})|) + n_{i(1)} \log(|\Sigma_{(1)}|) + y_{i(1)}^T (\Phi_{i(1)}(\tau_{(1)}) \otimes \Sigma_{(1)})^{-1} y_{i(1)} \right) \quad (6)$$

up to a constant. For the first two terms in the parentheses, we used the fact that if $\Sigma_i = \Phi \otimes \Sigma_{(1)}$, then $|\Sigma_i| = |\Phi|^p |\Sigma_{(1)}|^{n_{i(1)}}$. For the last term, note that with formula $(\Phi \otimes \Sigma)^{-1} = \Phi^{-1} \otimes \Sigma^{-1}$, it

can be rewritten as

$$\begin{aligned}
y_{i(1)}^T (\Phi_{i(1)} \otimes \Sigma_{(1)})^{-1} y_{i(1)} &= \text{vec}(Y_{i(1)}^T)^T (\Phi_{i(1)}^{-1} \otimes \Sigma_{(1)}^{-1}) \text{vec}(Y_{i(1)}^T) \\
&= \text{vec}(Y_{i(1)}^T)^T \text{vec}(\Sigma_{(1)}^{-1} Y_{i(1)}^T \Phi_{i(1)}^{-1}) \\
&= \text{tr}(Y_{i(1)} \Sigma_{(1)}^{-1} Y_{i(1)}^T \Phi_{i(1)}^{-1}) \\
&= \text{tr}(Y_{i(1)}^T \Phi_{i(1)}^{-1} Y_{i(1)} \Omega_{(1)}) \\
&\triangleq \text{tr} S_{i(1)} \Omega_{(1)},
\end{aligned} \tag{7}$$

where $S_{i(1)} = Y_{i(1)}^T \Phi_{i(1)}^{-1} Y_{i(1)}$. Here we also used the formula $\text{vec}(AXB) = (B^T \otimes A) \text{vec}(X)$ in line 2 and $\text{tr}(A^T B) = \text{vec}(A)^T \text{vec}(B)$ in line 3. Consequently, for log-likelihood function, we have

$$l_{(1)}(\Omega_{(1)}, \tau_{(1)} | y_{(1)}) = -\frac{1}{2} \left\{ \sum_{i=1}^m p \log(|\Phi_{i(1)}(\tau_{(1)})|) \right\} + \frac{1}{2} n_{(1)} \log(|\Omega_{(1)}|) - \frac{1}{2} n_{(1)} \text{tr}(\bar{S}_{(1)}(\tau_{(1)}) \Omega_{(1)}), \tag{8}$$

where $n_{(1)} = \sum_{i=1}^m n_{i(1)}$, $\bar{S}(\tau_{(1)}) = \frac{1}{n_{(1)}} \sum_{i=1}^m S_{i(1)}(\tau_{(1)})$. Here, we use $\bar{S}(\tau_{(1)})$ to emphasize that matrix $\bar{S}(\tau_{(1)})$ is a function of unknown parameter $\tau_{(1)}$.

Similarly, for the after treatment log-likelihood function, we have

$$l_{(2)}(\Omega_{(2)}, \tau_{(2)} | y_{(2)}) = -\frac{1}{2} \left\{ \sum_{i=1}^m p \log(|\Phi_{i(2)}(\tau_{(2)})|) \right\} + \frac{1}{2} n_{(2)} \log(|\Omega_{(2)}|) - \frac{1}{2} n_{(2)} \text{tr}(\bar{S}_{(2)}(\tau_{(2)}) \Omega_{(2)}), \tag{9}$$

where $n_{(2)} = \sum_{i=1}^m n_{i(2)}$, $\bar{S}(\tau_{(2)}) = \frac{1}{n_{(2)}} \sum_{i=1}^m S_{i(2)}(\tau_{(2)})$. Given above likelihood functions $l_{(1)}, l_{(2)}$, we define the estimates of Ω_1, Ω_2 through the following minimization problem,

$$\min_{\Omega_{(1)}, \Omega_{(2)}, \tau_{(1)}, \tau_{(2)}} \{ -2(l_{(1)}(\Omega_{(1)}, \tau_{(1)}) + l_{(2)}(\Omega_{(2)}, \tau_{(2)})) + P(\Omega_{(1)}, \Omega_{(2)}) \}$$

where $P(\cdot)$ is a convex penalty function. In the case of cross-sectional data, different versions of function $P(\cdot)$ have been proposed so that the resulting estimates of Ω_1 and Ω_2 could have some desirable properties such as sparsity, sharing common network structures. The following two penalty functions proposed in Danaher et al (2014) is adopted in the paper. The first penalty function, named as fused graphical lasso, is given by

$$P(\Omega_1, \Omega_2) = \lambda_1 \sum_{k=1}^2 \sum_{i \neq j} |\omega_{kij}| + \lambda_2 \sum_{ij} |\omega_{1ij} - \omega_{2ij}| \tag{10}$$

where if tuning parameter $\lambda_1 > 0$ is large, then the estimates of Ω_1 and Ω_2 will be sparse matrices while if tuning parameter $\lambda_2 > 0$ is large, then the estimates of Ω_1 and that of Ω_2 will bear more similarities. The second penalty, named as group graphical lasso, is given by

$$P(\Omega_1, \Omega_2) = \lambda_1 \sum_{k=1}^2 \sum_{i \neq j} |\omega_{kij}| + \lambda_2 \sum_{i \neq j} \sqrt{(\omega_{1ij}^2 + \omega_{2ij}^2)} \tag{11}$$

Here $\lambda_2 > 0$ encourage the same non-zero pattern for estimates of Ω_1 and Ω_2 .

Notice that given $\Omega_{(1)}, \Omega_{(2)}$, the objective function reduces to the following function of $\tau_{(1)}$ and $\tau_{(2)}$,

$$f(\tau_{(1)}, \tau_{(2)} | \Omega_{(1)}, \Omega_{(2)}) = \sum_{i=1}^m p(\log(|\Phi_{i(1)}|(\tau_{(1)})) + \log(|\Phi_{i(2)}|(\tau_{(2)}))) + n_{(1)} \text{tr}(\bar{S}_{(1)}(\tau_{(1)})\Omega_{(1)}) + n_{(2)} \text{tr}(\bar{S}_{(2)}(\tau_{(2)})\Omega_{(2)}) \quad (12)$$

The minimization of function $f(\cdot)$ with respect to $(\tau_{(1)}, \tau_{(2)})$ can be easily solved through e.g., Newton algorithms; while given $\tau_{(1)}, \tau_{(2)}$, the objective function reduces to convex function of $\Omega_{(1)}$ and $\Omega_{(2)}$,

$$g(\Omega_{(1)}, \Omega_{(2)} | \tau_{(1)}, \tau_{(2)}) = n_{(1)}(\log(|\Omega_{(1)}|) - \text{tr}(\bar{S}_{(1)}(\tau_{(1)})\Omega_{(1)})) + n_{(2)}(\log(|\Omega_{(2)}|) - \text{tr}(\bar{S}_{(2)}(\tau_{(2)})\Omega_{(2)})) + P(\Omega_1, \Omega_2) \quad (13)$$

Notice that $g(\cdot)$ is a convex function of $\Omega_{(1)}$ and $\Omega_{(2)}$. Efficient algorithms such as Alternating Direction Method of Multipliers (ADMM) have been proposed for minimizing $g(\Omega_{(1)}, \Omega_{(2)} | \tau_{(1)}, \tau_{(2)})$ with respect to $(\Omega_{(1)}, \Omega_{(2)})$. Given these observations, we recommend the following block coordinate descent procedure to solve the above problem.

Algorithm 1 Identify the networks based on homogeneous lglasso model

- 1: **procedure** GIVEN INITIAL VALUE $\Omega_{(1)}^{(0)} = \Omega_{(2)}^{(0)} = I_{p \times p}$, $\tau_{(1)}^{(0)}, \tau_{(2)}^{(0)}$, AND TUNING PARAMETERS λ_1, λ_2 AND ERROR TOLERANCE $e > 0$:
 - 2: At step $(k+1)$, the updated estimates $\hat{\tau}_{(1)}^{(k+1)}, \hat{\tau}_{(2)}^{(k+1)}$ of $(\tau_{(1)}, \tau_{(2)})$ are produced through minimizing $f(\tau_{(1)}, \tau_{(2)} | \Omega_{(1)}^{(k)}, \Omega_{(2)}^{(k)})$ with respect to $\tau_{(1)}, \tau_{(2)}$.
 - 3: With updated $\hat{\tau}_{(1)}^{(k+1)}, \hat{\tau}_{(2)}^{(k+1)}$, the updated estimates $(\hat{\Omega}_{(1)}^{(k+1)}, \hat{\Omega}_{(2)}^{(k+1)})$ of $(\Omega_{(1)}, \Omega_{(2)})$ are produced through minimizing $g(\Omega_{(1)}, \Omega_{(2)} | \hat{\tau}_{(1)}^{(k+1)}, \hat{\tau}_{(2)}^{(k+1)})$.
 - 4: **if** $|\hat{\tau}_{(1)}^{(k+1)} - \hat{\tau}_{(1)}^{(k)}| < e$, $|\hat{\tau}_{(2)}^{(k+1)} - \hat{\tau}_{(2)}^{(k)}| < e$, $|\hat{\Omega}_{(1)}^{(k+1)} - \Omega_{(1)}^{(k)}|_1 < e$ and $|\hat{\Omega}_{(2)}^{(k+1)} - \Omega_{(2)}^{(k)}|_1 < e$ **then**
 - 5: Stop and output the final estimates of $(\tau_{(1)}, \tau_{(2)}, \Omega_{(1)}, \Omega_{(2)})$ as $(\hat{\tau}_{(1)}^{(k+1)}, \hat{\tau}_{(2)}^{(k+1)}, \hat{\Omega}_{(1)}^{(k+1)}, \hat{\Omega}_{(2)}^{(k+1)})$.
 - 6: **else**
 - 7: return to Step 2.
 - 8: **end if**
 - 9: **end procedure**
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2.2. Heterogeneous models

In heterogeneous models, instead of assuming the same $\tau_{(1)}, \tau_{(2)}$ for all subjects, we assume subject-level correlation parameters $(\tau_{i(1)}, \tau_{i(2)})$ for each subject $i (1 \leq i \leq m)$. These parameters are independent random samplings from a appropriate two-dimensional distribution $(\tau_{i(1)}, \tau_{i(2)}) \sim \psi(x_1, x_2 | \alpha_1, \alpha_2, \theta)$. For example, ψ could be two-dimensional lognormal distribution. Another possible choice is the distribution that is constructed from copula function with exponential distribution as its marginals. In the latter case, we have $E\tau_{i(1)} = 1/\alpha_1$ and $E\tau_{i(2)} = 1/\alpha_2$ and parameter θ

measuring the correlation between $\tau_{i(1)}$ and $\tau_{i(2)}$, and we have the following joint distribution of $(y, \tau_{1(1)}, \tau_{1(2)}, \dots, \tau_{m(1)}, \tau_{m(2)})$,

$$\prod_{i=1}^m f(y_{i(1)}|\Omega_{(1)}, \tau_{i(1)})f(y_{i(2)}|\Omega_{(2)}, \tau_{i(2)})\psi(\tau_{i(1)}, \tau_{i(2)}|\alpha_1, \alpha_2, \theta), \quad (14)$$

For ease of exposition, we will use the independent copula generator in the following as an example to illustrate the proposed algorithms. Specifically, the joint density function in (14) can then reduce to

$$\prod_{i=1}^m f(y_{i(1)}|\Omega_{(1)}, \tau_{i(1)})f(y_{i(2)}|\Omega_{(2)}, \tau_{i(2)})\alpha_1\alpha_2 \exp(-\alpha_1\tau_{i(1)}) \exp(-\alpha_2\tau_{i(2)}), \quad (15)$$

from which the likelihood function is given by

$$\begin{aligned} L(\Omega_{(1)}, \Omega_{(2)}, \alpha_1, \alpha_2|y) &= \int_{\tau_1, \dots, \tau_m} \prod_{i=1}^m f(y_{i(1)}|\Omega_{(1)}, \tau_{i(1)}) \\ &\quad \times f(y_{i(2)}|\Omega_{(2)}, \tau_{i(2)})\alpha_1\alpha_2 \exp(-\alpha_1\tau_{i(1)}) \exp(-\alpha_2\tau_{i(2)})d\tau_{i(1)} \dots d\tau_{m(2)} \\ &= \int_{\tau_{1(1)}, \dots, \tau_{m(1)}} \prod_{i=1}^m f(y_{i(1)}|\Omega_{(1)}, \tau_{i(1)})\alpha_1 \exp(-\alpha_1\tau_{i(1)})d\tau_{i(1)} \dots d\tau_{m(1)} \\ &\quad \times \int_{\tau_{1(2)}, \dots, \tau_{m(2)}} \prod_{i=1}^m f(y_{i(2)}|\Omega_{(2)}, \tau_{i(2)})\alpha_2 \exp(-\alpha_2\tau_{i(2)})d\tau_{i(2)} \dots d\tau_{m(2)} \\ &\triangleq L_1(\Omega_{(1)}, \alpha_1|y_1)L_2(\Omega_{(2)}, \alpha_2|y_2) \end{aligned} \quad (16)$$

where

$$L_1(\Omega_{(1)}, \alpha_1|y_1) = \int_{\tau_{1(1)}, \dots, \tau_{m(1)}} \prod_{i=1}^m f(y_{i(1)}|\Omega_{(1)}, \tau_{i(1)})\alpha_1 \exp(-\alpha_1\tau_{i(1)})d\tau_{i(1)} \dots d\tau_{m(1)} \quad (17)$$

$$L_2(\Omega_{(2)}, \alpha_2|y_2) = \int_{\tau_{1(2)}, \dots, \tau_{m(2)}} \prod_{i=1}^m f(y_{i(2)}|\Omega_{(2)}, \tau_{i(2)})\alpha_2 \exp(-\alpha_2\tau_{i(2)})d\tau_{i(2)} \dots d\tau_{m(2)} \quad (18)$$

The sparse estimates for the network can then be obtained by minimizing the following penalized negative log-likelihood function,

$$\min_{\Omega_{(1)}, \Omega_{(2)}, \alpha_1, \alpha_2} \{-2(l_1(\Omega_{(1)}, \alpha_1|y_1) + l_2(\Omega_{(2)}, \alpha_2|y_2)) + P(\Omega_{(1)}, \Omega_{(2)})\}, \quad (19)$$

where $l_1(\Omega_{(1)}, \alpha_1|y_1) = \log(L_1(\Omega_{(1)}, \alpha_1|y_1))$, $l_2(\Omega_{(2)}, \alpha_2|y_2) = \log(L_2(\Omega_{(2)}, \alpha_2|y_2))$. Since no closed-form expressions for the integration in (17)-(18) are available, we propose Expectation-Maximization (EM) algorithm to find the solution to (19). Since we are considering the negative log-likelihood function in (19), the maximization in the EM algorithm will be replaced by minimization. The correlation parameters $\tau_{(1)} = (\tau_{1(1)}, \dots, \tau_{m(1)})$ and $\tau_{(2)} = (\tau_{1(2)}, \dots, \tau_{m(2)})$ are taken as the so-called missing data. Recall in the first step of the EM algorithm, the conditional distribution

of missing data $\tau = (\tau_{(1)}, \tau_{(2)})$ given $y, \Omega_{(1)} = \Omega_{(1)}^{(0)}, \Omega_{(2)} = \Omega_{(2)}^{(0)}, \alpha_1 = \alpha_1^{(0)}, \alpha_2 = \alpha_2^{(0)}$ has to be derived which can be shown to be

$$q(\tau|y, \Omega_{(1)}^{(0)}, \Omega_{(2)}^{(0)}, \alpha_{(1)}^{(0)}, \alpha_{(2)}^{(0)}) = \prod_{i=1}^m \prod_{j=1}^2 q(\tau_i|y_{i(j)}, \Omega_{(j)}, \alpha_j) \\ \propto \prod_{i=1}^m \prod_{j=1}^2 |\Phi_{i(j)}|^{-p/2} \exp(-y_{i(j)}^T (\Phi_{i(j)}^{-1} \otimes \Omega_{(j)}^{(0)}) y_{i(j)} / 2) \exp(-\alpha_{(j)}^{(0)} \tau_{i(j)}) \quad . \quad (20)$$

With (20) in hand, the expectation of the complete log-likelihood function for $(\mathbf{y}, \boldsymbol{\tau})$ with respect to (20) has to be computed. Given the joint density function (15) of $(\mathbf{y}, \boldsymbol{\tau})$, the expectation of its logarithmic transformation can be shown to be

$$Q(\Omega_1, \Omega_2, \alpha_1, \alpha_2 | \Omega_1^{(0)}, \Omega_2^{(0)}, \alpha_1^{(0)}, \alpha_2^{(0)}) = Q_1(\Omega_1, \alpha_1 | \Omega_1^{(0)}, \alpha_1^{(0)}) + Q_2(\Omega_2, \alpha_2 | \Omega_2^{(0)}, \alpha_2^{(0)}) \quad (21)$$

where

$$Q_1 = \sum_{i=1}^m p E_q \{\log(|\Phi_{i(1)}|)\} - n_1 \log(|\Omega_1|) + n_1 \text{tr} \left(S_{(1)}^{(0)} \Omega_1 \right) - 2m \log(\alpha_1) + 2\alpha_1 \sum_{i=1}^m E_g \tau_{i(1)} \\ Q_2 = \sum_{i=1}^m p E_q \{\log(|\Phi_{i(2)}|)\} - n_2 \log(|\Omega_2|) + n_2 \text{tr} \left(S_{(2)}^{(0)} \Omega_2 \right) - 2m \log(\alpha_2) + 2\alpha_2 \sum_{i=1}^m E_g \tau_{i(2)}$$

where

$$\bar{S}_{(1)}(\tau) = \frac{1}{n_1} \sum_{i=1}^m S_{i(1)}(\tau_i), \\ \bar{S}_{(2)}(\tau) = \frac{1}{n_2} \sum_{i=1}^m S_{i(2)}(\tau_i), \\ S_{(1)}^{(0)} = E_g \bar{S}_{(1)}(\tau) = \frac{1}{n} \sum_{i=1}^m E_q S_{i(1)}(\tau_i) \\ S_{(2)}^{(0)} = E_g \bar{S}_{(2)}(\tau) = \frac{1}{n} \sum_{i=1}^m E_q S_{i(2)}(\tau_i)$$

in which $S_{i(1)}(\tau_i)$ and $S_{i(2)}(\tau_i)$ are defined in (7) in Section 2.1. In the second step of the EM algorithm, the minimum point of the Q function in (21) has to be computed. As in 19, in order to generate sparse network estimates, the following penalized Q -function is minimized,

$$\min_{\Omega_1, \Omega_2, \alpha_1, \alpha_2} \{Q_1(\Omega_1, \alpha_1 | \Omega_1^{(0)}, \alpha_1^{(0)}) + Q_2(\Omega_2, \alpha_2 | \Omega_2^{(0)}, \alpha_2^{(0)}) + P(\Omega_1, \Omega_2)\} \quad (22)$$

where the penalty function $P(\cdot, \cdot)$ have the same form as in (10) or (11). For fixed (Ω_1, Ω_2) , it is straightforward to show that the minimum of the function in (22) with respect to (α_1, α_2) is attained at

$$\hat{\alpha}_1 = \frac{1}{\frac{1}{m} \sum_{i=1}^m E_q \tau_{i(1)}}, \quad \hat{\alpha}_2 = \frac{1}{\frac{1}{m} \sum_{i=1}^m E_q \tau_{i(2)}}, \quad (23)$$

i.e., the reciprocal of the sample mean of the conditional expectation of τ_i with respect to density (20). Then, for given (α_1, α_2) , the minimization of (22) with respect to (Ω_1, Ω_2) is equivalent to

$$\min_{\Omega_1, \Omega_2} \{-\log |\Omega_1| - \log |\Omega_2| + \text{tr}(S_{(1)}^{(0)} \Omega_1) + \text{tr}(S_{(2)}^{(0)} \Omega_2) + P(\Omega_1, \Omega_2)\}, \quad (24)$$

which will be solved through the ADMM algorithm as in (13). The difficult part of the minimization in (24) is to compute the expectation $E_q \tilde{S}_{(1)}(\tau)$ and $E_q \tilde{S}_{(2)}(\tau)$ involved in $S_{(1)}^{(0)}$ and $S_{(2)}^{(0)}$. which does not admit an analytical form given the complex form of density function $q(\tau)$ in (20); therefore, these expectations will be computed through Monte Carlo methods. This details of the algorithm are summarized in the table (2). Given initial value α_0 and Ω_0 , tuning parameter λ and error tolerance $e > 0$,

Algorithm 2 Identify the networks based on heterogeneous models

- 1: **procedure** GIVEN INITIAL VALUE $\alpha_0 = (\alpha_{10}, \alpha_{20})$ AND $\Omega_0 = (\Omega_{10}, \Omega_{20})$, TUNING PARAMETER λ_1, λ_2 AND ERROR TOLERANCE $e > 0$:
 - 2: Sample $\tau_{ij} = (\tau_{i(1)j}, \tau_{i(2)j}) (i = 1, \dots, m)$ from the distribution $q(\cdot)$ in (20) given the initial values of Ω 's and α 's;
 - 3: Estimate $E(\tau_{i(1)}), E(\tau_{i(2)}), E_g S_i(\tau_{i(1)}), E_g S_i(\tau_{i(2)}), S_{(1)}^{(0)}$ and $S_{(2)}^{(0)}$ by $\frac{1}{h} \sum_{j=1}^h \tau_{i(1)j}, \frac{1}{h} \sum_{j=1}^h \tau_{i(2)j}, \frac{1}{h} \sum_{j=1}^h S_i(\tau_{i(1)j}), \frac{1}{h} \sum_{j=1}^h S_i(\tau_{i(2)j}), \frac{1}{mh} \sum_{i=1}^m \sum_{j=1}^h S_i(\tau_{i(1)j})$ and $\frac{1}{mh} \sum_{i=1}^m \sum_{j=1}^h S_i(\tau_{i(2)j})$ respectively.
 - 4: Update α 's by $\hat{\alpha}$'s in (23), Ω 's by $\hat{\Omega}$'s, the solution to (24) in which $E(\tau)$'s and $S^{(0)}$'s are replaced by their estimates in Step 3.
 - 5: **if** $|\alpha_0 - \hat{\alpha}| < e$ and $|\Omega_0 - \hat{\Omega}|_1 < e$ **then**
 - 6: Stop and output $\hat{\alpha} = (\hat{\alpha}_1, \hat{\alpha}_2), \hat{\Omega} = (\hat{\Omega}_{(1)}, \hat{\Omega}_{(2)}), \hat{\tau}_i = (\hat{\tau}_{i(1)}, \hat{\tau}_{i(2)}), i = 1, \dots, m$.
 - 7: **else**
 - 8: $\alpha_0 = \hat{\alpha}$ and $\Omega_0 = \hat{\Omega}$, return to Step 2.
 - 9: **end if**
 - 10: **end procedure**
-

2.3. Selection of tuning parameters

For either homogeneous or heterogeneous models, in order to apply the models in practice, the tuning parameters (λ_1, λ_2) have to be specified. Cross validation has been proposed in literature for the network identification. However, cross validation is time consuming if the networks of interest are relatively high-dimensional. Here we propose an alternative way to select (λ_1, λ_2) , so-all extended quasi-likelihood information criterion (EQIC). EQIC can be seen as an extension of the conventional extended BIC (EBIC, ?). EBIC is likelihood-based method and was originally proposed in (Foygel and Drton (2010)) for identifying high-dimensional Gaussian graphical model from cross-sectional data. EQIC allows the likelihood in EBIC be replaced by a quasi likelihood when the computation of the likelihood is intractable. Specifically, EQIC has the following form for homogeneous model in Section 2.1,

$$EQIC = -2(l_1(\Omega_1) + l_2(\Omega_2)) + (|\Omega_1| + |\Omega_2|) \log(n) + 4\gamma(|\Omega_1| + |\Omega_2|) \log(p) \quad (25)$$

where l_1 and l_2 are defined in equations (8) and (9). If the data before and after the treatment is independent, then $l_1(\Omega_1) + l_2(\Omega_2)$ in the first term is just the log likelihood. Otherwise, it is not equal to the full likelihood and therefore we call it quasi likelihood. For heterogeneous models, where EM algorithm is employed to identify the networks, the terms $l_1(\Omega_1)$ and $l_2(\Omega_2)$ are replaced by the Q -function $Q_1(\Omega_1)$ and $Q_2(\Omega_2)$. With a little abuse of notations, we still call it EQIC methods. It should be noted that using the Q -function in EM algorithm for model selection has already been investigated extensively in literature, see, e.g., Ibrahim et al (2008).

3. Simulation study

In this section, the performance of the homogeneous/heterogeneous models in Section 2 are studied using simulated data. Specifically, we compare the homogeneous/heterogeneous models with two mainstream models, graphical lasso (Friedman et al (2008, 2019), GLASSO) and joint graphical lasso (Danaher et al (2014), JGL). The index used for measuring the performance is the so-called positive likelihood ratio (LR^+), which is defined as follows,

$$LR^+ = \frac{Sensitivity}{1 - Specificity} \quad (26)$$

where *Sensitivity* is the rate of the identified positives among the true positives, and *Specificity* is the rate of the identified negatives among the true negatives. Therefore, LR^+ measures how much more likely an edge is identified as an edge compared to a non-edge identified as an edge. The higher LR^+ , the more informative a positive result is. A value of 1 means the test results provide no diagnostic value.

3.1. Homogeneous model

First, let's consider the homogeneous models in Section 2.1. For the simulation, the parameters involved here include (1) the total number of subjects m , (2) correlation dampening rate τ , (3) average length of the data points per subject n_i , and (4) the number of nodes p . For m , we consider two choices, $m = 40, 80$. For τ , we consider three choices, $\tau = 0.3, 1, 3$. For n_i , we assume it follows a poisson distribution with two choices of mean value, $E(n_i) = 5, 10$. For p , we consider two choices, $p = 50, 100$. The density of edges in pre-vaccination networks is 20%. The post-vaccination network is generated by randomly flipping 10% of the connections of pre-treatment network into the opposites. This way the pre- and post-treatment networks share approximately 90% structures. For each of the combinations of above parameters and subject $i (1 \leq i \leq m)$, a $n_i \times p$ random matrix is generated. If the rows of the random matrix is stacked one after another to form a vector, then the distribution of this vector are multivariate normal with zero mean value $\mathbf{0}_{n_i p}$ and covariance matrix satisfying the equations (1)-(4). Here we set the parameter $k = 1$ in equations (2) and (4). With the simulated data in hand, the proposed homogeneous models are then applied to generate the estimates for both pre- and post-treatment networks with a given pair of tuning parameter (λ_1, λ_2) . The other two models, graphical lasso (GLASSO) and joint graphical lasso (JGL), are also carried out for the same simulated data. Specifically, for GLASSO, we ignore the correlations and apply the GLASSO to pre- and post-vaccination data separately with the same tuning parameter λ_1 . For JGL, the estimates of the pre- and post-treatment networks are obtained with the same form of penalty function and the same pair of tuning parameter (λ_1, λ_2) . It should be noted that ideally the choices of the sequence of (λ_1, λ_2) should be data-driven and vary with respect to the sample size m , the number of nodes p , the length of time points T *et al.* However, since the objective

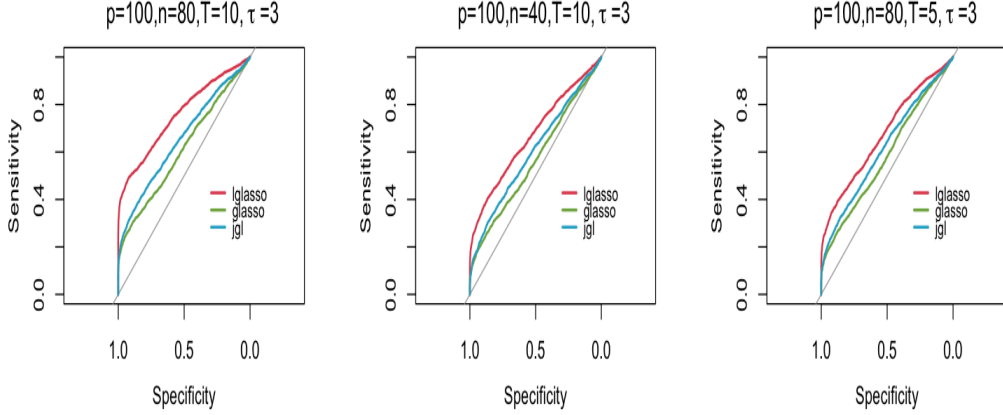


Figure 1: ROC curves of three network identification models where data are simulated from homogeneous model proposed in Section 2.1.

here is to compare the three models, we just use a fixed sequence of λ_1, λ_2 . Specifically, for all the scenarios mentioned above, tuning parameter λ_2 is fixed at 0.01, while the sequence for λ_1 is defined as $\{\exp(i_k), -8 \leq i_k \leq -2\}$ with i_k increasing equally from -8 to -2 for $1 \leq k \leq 100$. The results from these tuning parameters should reflect the systemic differences between the three models mentioned above.

In a given scenario, for each of the node pairs (a_i, a_j) , receiver operating characteristic (ROC) curve is used to measure which network best match the true network. To this end, we first calculate the empirical probability of nodes a_i, a_j being connected for each of the three models. It is defined as the proportion of networks in which a_i, a_j are connected among all candidate networks. Then with empirical probabilities in hand, the sensitivities and specificities in ROC curves can be calculated.

Three of these ROC curves are presented in Figure 1. In particular, the one on the left has most sample size $m = 80$ and time points $T = 10$. In all these cases, the proposed model *LGLASSO* always has higher sensitivities than both *GLASSO* and *JGL* for a given specificity, *i.e.*, *LGLASSO* outperforms *GLASSO* and *JGL*.

Figure 1 reflects the overall properties of all the candidate networks generated by *LGLASSO*. In practice, we need to select a specific network as the final estimate. Here we use EQIC proposed in equation (25) in Section 2.3 to select the optimal network for all three models. With the final network estimate in hand, LR^+ in equation (26) is used to measure the closeness of this network to the true underlying network. Table 1 lists the results for all the scenarios.

From Table 1, in all the cases, *LGLASSO* has the highest LR^+ among the three models. Furthermore, the LR^+ is a increasing function with respect to sample size n , time points T and τ . Particularly, since τ measures the rate by which the correlation between longitudinal data decreases, Table 1 shows that the correlation has a negative impact on the effectiveness of the network estimates for all three models. The proposed *LGLASSO* model has the highest performances which should be attributed to explicitly modeling of the correlations as opposed to other two models. Many of the LR^+ in Table 1 are just slightly higher than 1. This does not mean that those models have little power in that scenarios. This happened because we have used a fixed sequence of λ_1, λ_2 , where a better choice of tuning parameter sequence could have a much higher LR^+ .

		$p = 50$				$p = 100$			
		$m = 40$		$m = 80$		$m = 40$		$m = 80$	
		T=5	T=10	T=5	T=10	T=5	T=10	T=5	T=10
$\tau = 3$	lglasso	1.45	1.95	2.02	3.60	1.42	2.80	2.99	10.17
	glasso	1.19	1.31	1.41	1.71	1.12	1.25	1.44	1.73
	JGL	1.20	1.35	1.46	1.77	1.11	1.28	1.55	1.89
$\tau = 1$	lglasso	1.37	1.83	1.97	3.31	1.27	2.38	2.93	9.00
	glasso	1.13	1.16	1.26	1.35	1.04	1.08	1.26	1.27
	JGL	1.11	1.18	1.27	1.43	1.03	1.11	1.31	1.32
$\tau = 0.3$	lglasso	1.34	1.77	1.95	3.17	1.22	2.16	2.86	8.09
	glasso	1.12	1.10	1.21	1.25	1.01	1.04	1.17	1.17
	JGL	1.07	1.10	1.22	1.26	1.01	1.03	1.23	1.20

Table 1: Positive likelihood ratio LR^+ -based performance comparisons for homogeneous models. The tuning parameters are selected by EQIC.

3.2. Heterogeneous model

In this section, we consider the heterogeneous models. For heterogeneous models, we need to specify an exponential distribution for correlation dampening rate τ , i.e., $\tau \sim \exp(\alpha)$. For the simulation purpose, we consider three values for α , i.e., 0.3, 1 and 3. For other parameters, p, m, T and (λ_1, λ_2) , the same values will be used as in Section 3.1. Similar to Figure 1, Figure 2 presents the ROC curves that are computed from all the corresponding candidate networks. However, here ROC curves for the pre-treatment and post-treatment networks are plotted separately, from which we see that for both estimates of pre- and post-treatment network, the proposed LGLASSO models outperform the other two models. Furthermore, from Figure 2, the stronger the correlation between data at different time points (corresponding to $\alpha=3$), the more advantage of LGLASSO models have than the models GLSSSO and JGL.

In addition to the results generated from homogeneous models, heterogeneous models also generate random effects, $\tau_i (1 \leq i \leq m)$, for each subject. Figure 3 compares the estimated random effects $\hat{\tau}_i$ with the true τ_i . It can be seen that $\hat{\tau}_i$ is highly correlated to τ_i . Since in practice, τ_i is always unobservable. This allows us to treat $\hat{\tau}_i$ as a safe proxy for τ_i when the relationships between τ_i and other variables are of interest.

As for homogeneous models, we also use the EQIC to select the optimal network from the sequence of candidate networks generated from heterogeneous models. The subsequent LR^+ are computed and presented in Table 2. Note that since $E(\tau_i) = 1/\alpha$ for heterogeneous models, the cases with $\alpha = 0.3, 1, 3$ in Table 2 roughly correspond to the cases of $\tau = 3, 1, 0.3$ in Table 1. It is apparent that the proposed LGLASSO model always has the highest LR^+ among the three models in all the cases considered. Therefore, LGLASSO is the most powerful model in terms of identifying the structure of the true networks.

4. Real data analysis

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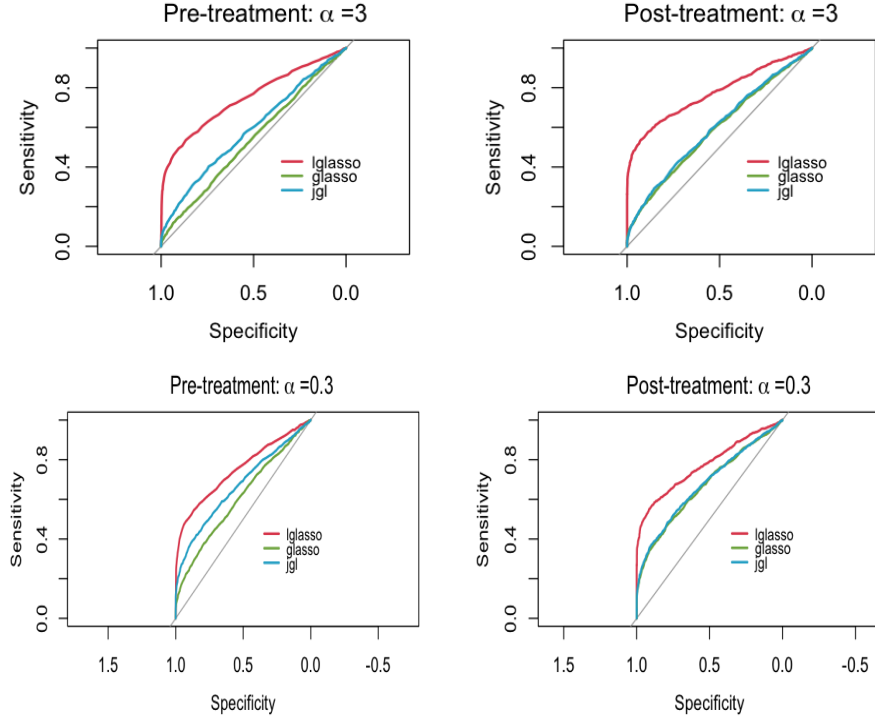


Figure 2: ROC curves of the network identification models when data is simulated from heterogeneous model in Section 2.2. Here $p = 100, n = 80, T = 10$ for all the four cases.

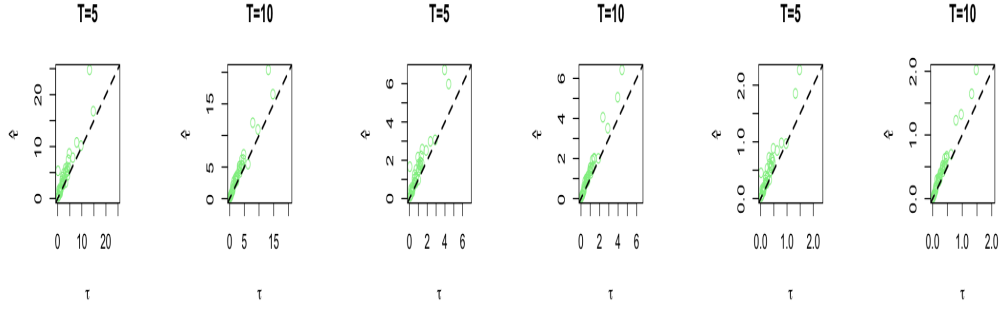


Figure 3: Estimated random effects vs true random effects corresponding to the $n = 40$ cases in Table 2. From top to bottom, the parameter α for the row is $\alpha = 0.3$, $\alpha = 1$ and $\alpha = 3$ respectively. The number of data points T for each subject is $E(T) = 5$ for the panels on the left and $E(T) = 10$ for the panels on the right.

		$p = 50$				$p = 100$			
		$m = 40$		$m = 80$		$m = 40$		$m = 80$	
		T=5	T=10	T=5	T=10	T=5	T=10	T=5	T=10
$\alpha = 0.3$	lglasso	1.47	2.21	2.03	3.60	1.27	1.81	2.02	3.77
	glasso	1.18	1.21	1.34	1.53	1.10	1.15	1.35	1.48
	JGL	1.19	1.25	1.39	1.56	1.12	1.17	1.41	1.59
$\alpha = 1$	lglasso	1.50	2.30	2.06	3.75	1.27	1.76	1.94	3.56
	glasso	1.12	1.14	1.25	1.34	1.05	1.08	1.23	1.20
	JGL	1.13	1.16	1.26	1.34	1.06	1.09	1.27	1.26
$\alpha = 3$	lglasso	1.51	2.33	2.10	3.81	1.25	1.74	1.88	3.49
	glasso	1.07	1.10	1.17	1.25	1.02	1.05	1.15	1.14
	JGL	1.07	1.12	1.20	1.27	1.05	1.05	1.20	1.16

Table 2: Positive likelihood ratio-based performance comparisons for heterogeneous models. The tuning parameters are selected by EQIC.

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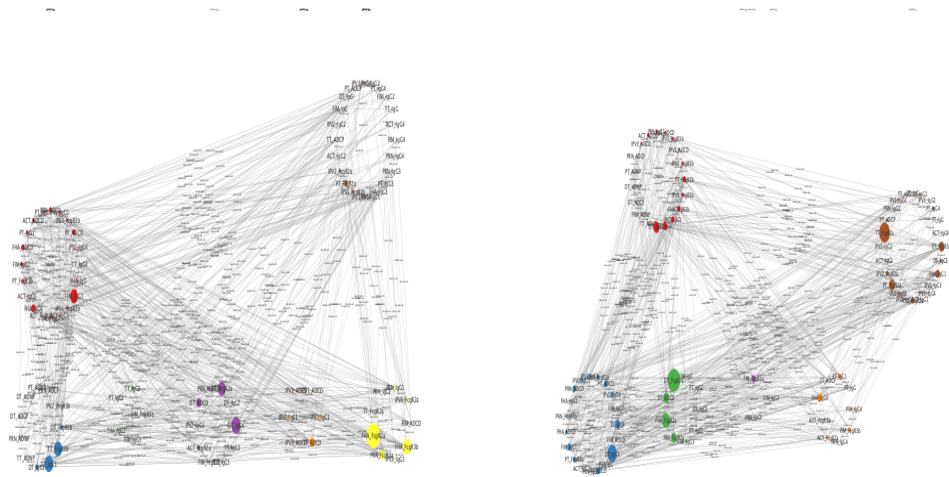


Figure 4: The estimated pre- and post-vaccination (from left to right) feature networks for infant immunity

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